

2. Statistics (Parameter Estimation)

- ➊ Introduction to Statistical Inference
- ➋ Point Estimation
 - Method of moments
 - Maximum likelihood estimation
 - Properties of MLE
- ➌ Interval Estimation (Confidence Interval)

Reading: Rice Ch7-8, 10.2-10.5; Casella and Berger Ch7, Ch9, 10.1, 10.4.

Introduction to Statistical Inference

Data: $x_1, \dots, x_n$

population

$X \sim F(x|\theta)$

θ unknown parameter.

$$\hat{\theta} = g(x_1, \dots, x_n) \Rightarrow \theta$$

Given a random (i.i.d.) sample $X_1, X_2, \dots, X_n \sim F(x)$, where population distribution $F(x)$ is often unknown,

Question: What can we infer about $F(x)$?

Statistical inference

- ① **Non-parametric model:** $F(x)$ is completely unknown.
- ② **Parametric model:** the form of $F(x)$ is known, but depending on some unknown parameters θ . $\theta = (\theta_1, \dots, \theta_p)$

- Estimation
- Hypothesis testing

$H_0: \mu = \mu_0$ whether can be reject.

$$X_i \sim N(\mu, \sigma^2), \quad \theta = (\mu, \sigma^2).$$

Basic Concepts

- Parameter:** The fixed but unknown constant(s), i.e., θ in the joint distribution of $X_1, \dots, X_n$ are called parameters. Sometimes, functions of θ are also called parameters.
- Estimator:** A (point) estimator, denoted by $\hat{\theta}$, of a parameter θ is a **statistic** used to estimate θ and a (point) **estimate** is a **value** of $\hat{\theta}$ computed based on the observed data $x_1, \dots, x_n$.

$$\hat{\theta} = g(x_1, \dots, x_n)$$
- Note.** An estimator is a random variable and an estimate is a number.

Example 3.1: We measure the heights of a sample of 5 students in our class, we may record heights 178, 182, 170, 161, 164. ✓
 Then

- what does X_1, X_2, X_3, X_4, X_5 stand for? Summarize $x_1 \dots x_n \Rightarrow \hat{\mu} = ?$
 - If one assumes $X_i \sim N(\mu, 1)$ and uses sample mean $\bar{X} = \frac{X_1 + X_2 + X_3 + X_4}{4}$ to estimate μ , are $\bar{X}$ and $\bar{x} = 171$ the same?
- estimator* $\bar{X}$ *estimate of μ*

Summary Notes

Distinction between $X_1, \dots, X_n$ and $x_1, \dots, x_n$

- $X_1, \dots, X_n$ are random variables while $x_1, \dots, x_n$ are real values.
- Different time points: Before the data are collected $\Rightarrow X_1, \dots, X_n$;
After the data collected $\Rightarrow x_1, \dots, x_n$.
- Statistical inference is developed based on $X_1, \dots, X_n$, instead of $x_1, \dots, x_n$. That is, statistical procedure is suitable for all possible (future) observations under the consideration of their uncertainty, not only for the currently (past) observed data.

Distinction between θ and $\hat{\theta}$

- Parameter θ is a **fixed** unknown quantity, while estimator $\hat{\theta}$ is a statistic (function of random sample) so it is a **random**.

$$\hat{\theta}(x_1, \dots, x_n) \xrightarrow{P} \theta$$

$n \rightarrow \infty$

Point Estimation

Estimation: A procedure to find an estimate based on a sample that is close to the unknown parameter θ of interest.

Point estimate: a single value or point for $\hat{\theta}$

- **Bias** of an estimator $\hat{\theta}$ is defined by



$$\text{bias}(\hat{\theta}) = E(\hat{\theta}) - \theta.$$

random $\hat{\theta}$ $\rightarrow$ fixed $\hat{\theta}$

$$E(\hat{\theta} - \theta) = E(\hat{\theta}) - \theta$$

$\text{bias}(\hat{\theta}) = 0.$

- $\hat{\theta}$ is called an unbiased estimator if $E(\hat{\theta}) = \theta$. Otherwise $\hat{\theta}$ is said to be biased.
- Standard error (SE): The standard error of an estimator is the standard deviation of its sampling distribution, i.e.,

$$\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}.$$

Note. SE depends on the unknown distribution $F(x)$, so is an unknown quantity but can be estimated.

For any point estimator, both bias and SE has to be considered to justify its quality.

- Mean squared error (MSE): The mean squared error of a point estimator $\hat{\theta}$ is defined by

$$\underline{MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2.} \quad \underline{E(\hat{\theta} - \theta) \Rightarrow \text{bias}}$$

The MSE can be computed by

$$\underline{MSE(\hat{\theta}) = [\text{bias}(\hat{\theta})]^2 + \text{Var}(\hat{\theta}).}$$

Why? $E(\hat{\theta} - \theta)^2 = E(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2$

$$= \underbrace{E(\hat{\theta} - E(\hat{\theta}))^2}_{\text{Var } \hat{\theta}} + \underbrace{E(E(\hat{\theta}) - \theta)^2}_{\text{const}} + 2 \underbrace{E(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)}_{=0}$$

$$= \text{Var } \hat{\theta} + \underbrace{(E(\hat{\theta}) - \theta)^2}_{\text{bias}(\hat{\theta})^2}$$

$$h(x) = \frac{x(1-x)}{n}$$

$$E(X_i) = p.$$

- **Example 3.2:** Let $X_1, \dots, X_n$ follow Bernoulli(p) distribution i.i.d. and $\hat{p} = \sum_{i=1}^n X_i / n$. Then

estimator of p .

$$E(\hat{p}) = \sum_{i=1}^n E(X_i) / n = p.$$

$$\text{Var}(\hat{p}) = \text{Var}(\bar{X}) = \frac{\text{Var} X_i}{n} = \frac{p \cdot (1-p)}{n}$$

So $\hat{p}$ is an unbiased estimator of p . The standard error is

$$SE(\hat{p}) = \sqrt{\text{Var}(\hat{p})} = \sqrt{p(1-p)/n}, \quad \text{unknown.}$$

then the estimated SE, $\hat{SE}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$.

Now suppose we have 10 observations which contains 6 "1"s and 4 "0"s. Estimates $\hat{p} = ?$ and $\hat{SE}(\hat{p}) = ?$

- **Example 3.1 (cont.):** $X_i \sim N(\mu, 1)$, we use $\hat{\mu} = \bar{X} = \frac{X_1 + X_2 + X_3 + X_4}{4}$. Is $\hat{\mu}$ unbiased?

What is the estimate of μ given 5 observations?

Find the estimated SE of $\hat{\mu}$.

$$SE(\hat{\mu}) = SE(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{1}{n}}$$

- **Example 3.3:** Let $X_1, \dots, X_n$ be a random sample from population (μ, σ^2) (not necessary to be normal). Show that sample variance

$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator for σ^2 and

$S^{*2} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is a biased estimator.

$ES^2 = \sigma^2$, bias = $ES^2 - \sigma^2 = 0$. unbiased.

$E \sum_{i=1}^n (X_i - \bar{X})^2$

$= E \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2$

$= E \sum_{i=1}^n [(X_i - \mu)^2 + (\mu - \bar{X})^2 + 2(X_i - \mu)(\mu - \bar{X})]$

$= \sum_{i=1}^n [E(X_i - \mu)^2 + E(\mu - \bar{X})^2 + 2E(X_i - \mu)(\mu - \bar{X})]$

$= n\sigma^2 + n \cdot \sigma^2/n - 2n \cdot \sigma^2/n$

$= (n-1)\sigma^2$

$ES^2 = \sigma^2 \checkmark$

$ES^{*2} = \left(\frac{n-1}{n}\right) \sigma^2$

$(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$

$E[(n-1)S^2/\sigma^2] = n-1$

mean of $\chi^2_n = n$

under condition:
 $X_i \sim N(\mu, \sigma^2)$

$E\bar{X} = \mu$

$\text{Var } \bar{X} = \frac{\text{Var } X_i}{n} = \frac{\sigma^2}{n}$

$= E(\bar{X} - E\bar{X})^2 = E(\bar{X} - \mu)^2$

$2 \sum_{i=1}^n E(X_i - \mu) \cdot (\mu - \bar{X})$
 $= 2 E \left(\sum_{i=1}^n (X_i - \mu) \cdot (\mu - \bar{X}) \right) = 2n E(\mu - \bar{X})$
 $(n\bar{X} - n\mu)$

To summarize, we estimate θ using some statistic $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$, where $X_1, \dots, X_n$ is a random sample from population $F(x)$.

Sensible criteria to evaluate $\hat{\theta}$

- Small SE
- Small bias
- Small MSE
- Consistent $\hat{\theta}(X_1, \dots, X_n) \rightarrow \theta$ in probability, as $n \rightarrow \infty$.
- Robust (gives a good estimate, even if some bad data present in the sample)

Example 3.4. Both the sample mean $\bar{X}$ and sample variance S^2 are consistent and unbiased estimators of the population (underlying distribution) mean and variance.

How can we find a good point estimator for θ ?

The Method of Moments

$$F(x|\theta) \Rightarrow \overbrace{E(X), E(X^2), \dots, E(X^k)}^{\text{Sample Data: } x_1, \dots, x_n} \Rightarrow \underbrace{\frac{1}{n} \sum_{i=1}^n x_i}_{M_1(\vec{X})}, \underbrace{\frac{1}{n} \sum_{i=1}^n x_i^2}_{M_2(\vec{X})}, \dots, \underbrace{\frac{1}{n} \sum_{i=1}^n x_i^k}_{M_k(\vec{X})}$$

Idea: Assume that sample moments should provide good estimates of the corresponding population moments.

- Random variable X follows a distribution. The k^{th} moment of the distribution, if exists, is

$$\mu_k = E(X^k).$$

- If $X_1, \dots, X_n$ is a random sample from a distribution, the k^{th} sample moment is defined by

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k.$$

- μ_k is a function of population parameters.

$$f(x) = \lambda e^{-\lambda x}$$

$$F(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$$

Method of moments estimator (MME) of the parameter(s) is obtained by solving the equation(s)

$$\mu_k(\theta) = m_k(X_i)$$

$$\mu_k = m_k, \quad \text{for } k = 1, \dots, K,$$

where K is the number of parameters to be estimated.

- **Example 3.5.** Suppose $X_1, \dots, X_n$ are i.i.d. $\sim \text{Pois}(\lambda)$. The first moment of $\text{Pois}(\lambda)$ is λ , while the first sample moment is $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

$$E(X) = \lambda$$

$$\mu_1 = m_1 \Rightarrow \lambda = \bar{X}$$

$\Rightarrow$ The MME of λ is $\hat{\lambda} = \bar{X}$.

As an example, the number of customers arrived in during 10am-12pm were recorded for 23 days:

31, 29, 19, 18, 31, 28, 34, 27, 34, 30, 16, 18

26, 27, 27, 18, 24, 22, 28, 24, 21, 17, 24.

MM estimate of λ is then $\hat{\lambda} = 24.9$. $SE(\hat{\lambda}) = ?$

$$i = 1, \dots, n$$

$$\text{E.g. } X_i \sim N(\mu, \sigma^2),$$

$$\mu_1 = m_1, \quad \hat{\mu} = \bar{X}$$

$$\mu_2 = m_2$$

$$E(X^2) = \text{Var}(X) + (E(X))^2$$

$$= \sigma^2 + \mu^2$$

$$\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n} \sum X_i^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum X_i^2 - \bar{X}^2 = S^{*2}$$

$$x \sim U(a, b) \quad \text{pdf } f(x) = \frac{1}{b-a}$$

- **Example 3.6.** $X_1, \dots, X_n \sim \text{Uniform}(\underline{0}, \theta)$ i.i.d., parameter θ is unknown. Use the method of moments to estimate θ .

by MM. $\mu_1 = m_1$ $E X_i = \theta/2$ $m_1 = \frac{1}{n} \sum X_i$

$$\frac{\theta}{2} = \bar{X}$$

$$\hat{\theta} = 2\bar{X}$$

$$E(\hat{\theta}) = 2 \cdot E(\bar{X}) = 2 \cdot \theta/2 = \theta$$

unbiased,

$$SE(\hat{\theta}) = \sqrt{\text{Var}(2\bar{X})} = \sqrt{4 \frac{\text{Var} X_i}{n}} = \sqrt{4 \cdot \frac{\theta^2}{12n}} = \frac{2\theta}{\sqrt{3n}}$$

$$\hat{SE}(\hat{\theta}) = 2\bar{X}/\sqrt{3n}$$

$$= \theta/\sqrt{3n} \quad \text{as } n \rightarrow \infty \quad SE \rightarrow 0$$

- **Exercise 3.1.** $X_1, \dots, X_n \sim \text{Gamma}(\alpha, \lambda)$ i.i.d., parameters α and λ are unknown. Find their estimator using the method of moments.

$$\mu_1 = E X_i = \alpha/\lambda \quad , \quad m_1 = m_1 \Rightarrow \alpha/\lambda = \bar{X}$$

$$\mu_2 = E X_i^2 = \text{Var}(X_i) + (E X_i)^2, \quad m_2 = m_2 \Rightarrow \frac{\alpha}{\lambda^2} + \frac{\alpha^2}{\lambda^2} = \frac{1}{n} \sum X_i^2$$

$$= \alpha/\lambda^2 + \alpha^2/\lambda^2$$

Maximum Likelihood Estimation

The likelihood function is one of the main tools in statistical inference, allowing the data to provide information about a statistical model.

Key idea: Assume that a probability distribution (model) is appropriate for our data. Then the most likely parameter should provide the largest "probability" of obtaining a particular observation with the given model.

$$\max_{\theta} P(\text{observing data } x_1, \dots, x_n \mid X_i \sim F(x|\theta))$$

- **Likelihood function** $L(\theta|\mathbf{x})$ of parameter θ given observations $\mathbf{x} = (x_1, \dots, x_n)$ of random sample $X_1, \dots, X_n$ is defined by

- for discrete X_i with pmf $p(x_i|\theta)$

$P(X_i = x_i) = \text{pmf at } x_i$

observing x_i , $\varepsilon > 0$, small

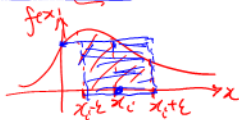
$X_i \in (x_i - \varepsilon, x_i + \varepsilon)$

$P(X_i \in (x_i - \varepsilon, x_i + \varepsilon))$

$$= \int_{x_i - \varepsilon}^{x_i + \varepsilon} f(x) dx \approx f(x_i) \cdot 2\varepsilon$$

- for continuous X_i with pdf $f(x_i|\theta)$

$$L(\theta|\mathbf{x}) = f(x_1|\theta) \times \dots \times f(x_n|\theta)$$



$$\prod_{i=1}^n P(X_i \in (x_i - \varepsilon, x_i + \varepsilon)) \approx \left(\prod_{i=1}^n f(x_i|\theta) \right) (2\varepsilon)^n$$

- Maximum likelihood estimator (MLE) is $\hat{\theta} = \hat{\theta}(\mathbf{X})$ that maximizes the likelihood function given sample $\mathbf{X} = (X_1, \dots, X_n)$. That is

$$\hat{\theta} = \arg \max_{\theta} [L(\theta|\mathbf{X})] = \arg \max_{\theta} [l(\theta|\mathbf{X})],$$

where $l(\theta|\mathbf{X}) = \log L(\theta|\mathbf{X}) = \sum_{i=1}^n \log f(X_i|\theta)$.

- Note:** Typical approach to function maximization problem
 - Differentiate $l(\theta|\mathbf{x})$ wrt θ and let it equal to 0
 - Check root is maximum, not minimum or saddle point.

$$\frac{\partial l(\theta)}{\partial \theta} = 0 \Rightarrow \hat{\theta}. \text{ MLE.}$$

$$\frac{\partial^2 l(\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}} < 0.$$

$\hat{\theta}$ is the maximum point



data $N(\mu, \sigma^2)$ $\rightarrow$ Sample $n=25$
 $\hat{\mu}, \hat{\sigma}^2$

Example 3.7. An exploratory data analysis of a sample of cholesterol measurements from a group of patients suggests that the normal distribution might be a good model for cholesterol levels for people with this condition. X is the cholesterol level of any new patient with the same condition, then $X \sim N(\mu, \sigma^2)$. Now suppose the data consist of the following 25 measurements (units are mmol/L):

$\left\{ \begin{array}{l} 4.57 \ 5.20 \ 6.16 \ 6.20 \ 5.33 \ 5.57 \ 4.43 \ 5.31 \ 5.72 \ 6.36 \\ 4.28 \ 4.71 \ 4.23 \ 5.48 \ 6.09 \ 6.11 \ 5.80 \ 5.50 \ 3.99 \ 5.41 \\ 5.90 \ 5.81 \ 6.17 \ 5.59 \ 5.65 \end{array} \right\}$ Sample

$\vec{x} = (x_1, \dots, x_n), n=25$

Find the ML estimates of parameters μ and σ^2 .

$$L(\mu, \sigma^2 | \vec{x}) = \prod_{i=1}^n f(x_i | \mu, \sigma^2), \quad \text{pdf } f(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$= (2\pi\sigma^2)^{-\frac{n}{2}} \cdot \exp\left\{-\frac{\sum (x_i - \mu)^2}{2\sigma^2}\right\}$$

$$\ell \hat{=} \log L(\mu, \sigma^2 | \vec{x}) = -\frac{n}{2} (\log(2\pi) + \log \sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{\partial \ell}{\partial \mu} = 0 \Rightarrow \frac{1}{2\sigma^2} \cdot 2 \sum (x_i - \mu) = 0 \Rightarrow \hat{\mu} = \frac{1}{n} \sum x_i = \bar{x}$$

$$\frac{\partial \ell}{\partial \sigma^2} = 0 \Rightarrow -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = S^{*2}$$

Example 3.8. Given sample data $x_1, \dots, x_n$, find the MLEs for the parameters of a $\text{Gamma}(\alpha, \lambda)$ distribution with pdf

$$f(x|\alpha, \lambda) = \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x}.$$

Soln. The log-likelihood function

$$\begin{aligned} \underline{l(\alpha, \lambda | \mathbf{x})} &= \underline{\sum_{i=1}^n \log f(x_i | \alpha, \lambda)} \\ &= \underline{n\alpha \log \lambda + (\alpha - 1) \sum_{i=1}^n \log x_i - \lambda \sum_{i=1}^n x_i - n \log \Gamma(\alpha)} \end{aligned}$$

The ML estimating equations are

$$\begin{aligned} 0 &= \underline{\frac{dl}{d\alpha}} = n \log \lambda + \sum_{i=1}^n \log x_i - n \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} \\ 0 &= \underline{\frac{dl}{d\lambda}} = \frac{n\alpha}{\lambda} - \sum_{i=1}^n x_i \end{aligned}$$

Solving both equations, we have MLEs satisfy

$$\left. \begin{aligned} \hat{\lambda} &= \frac{\hat{\alpha}}{\bar{x}} \\ 0 &= n \log \hat{\alpha} - n \log \bar{x} + \sum_{i=1}^n \log x_i - n \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} \end{aligned} \right\}$$

No explicit solution!

A numerical iteration method (e.g., Newton Raphson algorithm) can be applied to find an approximate solution.

$\hat{\alpha}, \hat{\lambda}$, $x_1, \dots, x_n$, estimate

$$X_i = \begin{pmatrix} 1, \theta \\ 0, 1-\theta \end{pmatrix} \quad \text{pmf.}$$

Exercise 3.2. $X_1, \dots, X_n$ is a random sample from a *Bernoulli*(θ)

distribution with pmf $p(x|\theta) = \begin{cases} \theta^x(1-\theta)^{1-x}, & \text{if } x=0 \text{ or } 1 \\ 0, & \text{otherwise} \end{cases}$,

$0 \leq \theta \leq 1$. Find the MLE of θ .

$$L(\theta|\vec{x}) = \prod_{i=1}^n p(x_i|\theta) = \theta^{\sum x_i} (1-\theta)^{\sum (1-x_i)} = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$$

$$l(\theta|\vec{x}) = \log L = \sum x_i \cdot \log \theta + (n - \sum x_i) \log(1-\theta).$$

$$\frac{dl}{d\theta} = 0 \Rightarrow \sum x_i \cdot \frac{1}{\theta} + (n - \sum x_i) \frac{-1}{1-\theta} = 0 \Rightarrow \sum x_i - (n - \sum x_i) = 0 \Rightarrow \hat{\theta} = \bar{x}$$

Exercise 3.3. (Laplace distribution) Let $X_1, \dots, X_n$, be i.i.d. with pdf



$$f(x|\theta) = \frac{1}{2} e^{-|x-\theta|}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty.$$

Find the MLE of θ .

$$L(\theta|\vec{x}) = \prod_{i=1}^n f(x_i|\theta) = \frac{1}{2^n} e^{-\sum |x_i - \theta|} \quad \text{sgn}(x_i - \theta)$$

$$l(\theta|\vec{x}) = \log L = -n \log 2 - \sum_{i=1}^n |x_i - \theta|.$$

$$\frac{dl}{d\theta} = \sum_{i=1}^n \text{sgn}(\theta - x_i) = 0. \quad \hat{\theta} = \text{median}(X_1, \dots, X_n)$$

$$\frac{d|x_i - \theta|}{d\theta} = \begin{cases} -1, & x_i > \theta \\ 0, & x_i = \theta \\ 1, & x_i < \theta \end{cases}$$

$$\underline{X_{(1)}} \leq X_{(2)} \leq \dots \leq \underline{X_{(n)}}.$$

Example 3.9. (Uniform distribution) Let $X_1, \dots, X_n$ be i.i.d. from

$$\text{Uniform}(0, \theta) \text{ with pdf } f(x|\theta) = \begin{cases} \frac{1}{\theta}, & 0 < x \leq \theta \\ 0, & \text{otherwise} \end{cases} = \frac{1}{\theta} I(0 < x \leq \theta)$$

Find the MLE of θ .

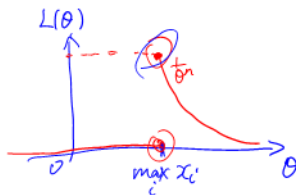
$$\begin{aligned} \underline{L(\theta|\vec{x})} &= \prod_{i=1}^n f(x_i|\theta) = \frac{1}{\theta^n} \prod_{i=1}^n I(0 < x_i \leq \theta). \\ &= \frac{1}{\theta^n} \cdot I(0 < \max_i x_i \leq \theta). \end{aligned}$$

indicator function

$I(A) = \begin{cases} 1 & A \text{ is true} \\ 0 & \text{others} \end{cases}$

$0 < x_1 \leq \theta, \dots, 0 < x_n \leq \theta.$

$$\hat{\theta} = \max_i x_i = \underline{X_{(n)}}.$$



≡

Ex. $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ i.i.d. MLE of λ .

$$L(\lambda | \vec{x}) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i} \quad \ell(\lambda | \vec{x}) = n \log \lambda - \lambda \sum x_i$$

Properties of MLE

$$E(\hat{\lambda}) = E(\frac{1}{\hat{x}})$$

$$\text{Var}(\hat{\lambda})$$

$$\text{AVar}(\hat{\lambda})$$

$$\frac{\partial \ell}{\partial \lambda} = 0 \Rightarrow \frac{n}{\lambda} - \sum x_i = 0 \Rightarrow \hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$$

$$\text{MLE of } \mu = 1/\lambda \Rightarrow \hat{\mu} = 1/\hat{\lambda}$$

$$\ell(\mu | \vec{x}) = -n \log \mu - \frac{1}{\mu} \sum x_i$$

- Invariance:** Suppose that θ is the parameter associated with a distribution. $g(\theta)$ is a function of θ . If $\hat{\theta}$ is the MLE of θ , then the MLE of $g(\theta)$ is given by $g(\hat{\theta})$.

$$\text{MLE } \hat{\theta} \quad \text{Any function of } \theta. \quad g(\theta)$$

$$\text{MLE of } g(\theta) : g(\hat{\theta})$$

$$\theta = \hat{\theta}$$

$$SE(\hat{\theta}) = g'(\theta)$$

$$SE(\hat{\theta}) = g'(\hat{\theta})$$

$$\Rightarrow \text{MLE of } SE(\hat{\theta})$$

$$\max_{\theta} L(\theta | \vec{x}) = \max_{g(\theta)} L^*(g(\theta) | \vec{x})$$

$$\text{when } \theta = \hat{\theta}, L^*(g(\theta) | \vec{x}) \text{ achieves its max. } \Rightarrow g(\theta) = g(\hat{\theta})$$

- Consistency:** Under appropriate smoothness (regularity) conditions on pdf f , the MLE $\hat{\theta}$ is consistent (by LLN).

θ shouldn't be on the boundary of domain of $f(x|\theta)$.

$$\hat{\theta} \xrightarrow{P} \theta \text{ as } n \rightarrow \infty.$$

$$\hat{\theta} \approx \theta$$

$$I_{x_1, \dots, x_n}(\theta) = n I(\theta)$$

$$= -E\left(\frac{d^2}{d\theta^2} \log f(\vec{x}|\theta)\right), \quad \log f(\vec{x}|\theta) = \sum_{i=1}^n \log f(x_i|\theta)$$

$$a = \sqrt{n I(\theta_0)}, \quad b = -\sqrt{n I(\theta_0)} \cdot \theta_0$$

$$a\hat{\theta} + b \overset{a}{\sim} N(0, 1) \quad E(a\hat{\theta} + b) = 0$$

$$\hat{\theta} \overset{a}{\sim} N(\theta_0, \frac{1}{n I(\theta_0)}) \quad aE\hat{\theta} = -b$$

$$E\hat{\theta} \approx \theta_0 \quad \text{Var } a\hat{\theta} = 1 \quad \text{Var } \hat{\theta} = \frac{1}{a^2}$$

- **Asymptotic Normality:** Under appropriate smoothness conditions on f , the probability distribution of $\sqrt{n I(\theta_0)}(\hat{\theta} - \theta_0)$ tends to standard normal (by CLT). That is,

$$\frac{\log f(\vec{x}|\hat{\theta})}{\log f(\vec{x}|\theta_0)}$$

$$\sqrt{n I(\theta_0)}(\hat{\theta} - \theta_0) \sim N(0, 1)$$

const $\uparrow$ r.v. $\uparrow$ const

approximately for large n , where θ_0 is the true value of parameter θ . $I(\theta)$ is the Fisher information defined by $\log f(\hat{\theta}|\vec{x}) = \sum_{i=1}^n \log f(x_i|\theta)$.

$$I(\theta) = E\left[\left(\frac{\partial}{\partial \theta} \log f(X|\theta)\right)^2\right] = -E\left[\frac{\partial^2}{\partial \theta^2} \log f(X|\theta)\right]$$

log like for a single data point

E.g.

Exp(x)

$$\text{Var}(\hat{\lambda}) = \frac{1}{n I(\lambda)} = \frac{\lambda^2}{n}$$

$$I(\lambda) = -E\left(\frac{d^2}{d\lambda^2} \log f(x|\lambda)\right) = -E\left(\frac{d}{d\lambda} \left(\frac{1}{\lambda} - x\right)\right)$$

$$= -E\left(-\frac{1}{\lambda^2}\right) = 1/\lambda^2$$

Note. The MLE is approximately unbiased and its MSE is approximately $\frac{1}{n I(\theta_0)}$.

$$\text{MSE}(\hat{\theta}) = \text{bias}^2(\hat{\theta}) + \text{Var}(\hat{\theta})$$

$$\approx 0 + \text{Var}(\hat{\theta})$$

$$= \frac{1}{n I(\theta_0)}$$

Efficiency: A way to compare two estimators having the same bias. The efficiency of $\hat{\theta}$ relative to $\tilde{\theta}$ is

$$\text{eff}(\hat{\theta}, \tilde{\theta}) = \frac{\text{Var}(\tilde{\theta})}{\text{Var}(\hat{\theta})}$$

$\text{Var}(\hat{\theta}) < \text{Var}(\tilde{\theta})$
 $\text{eff}(\hat{\theta}, \tilde{\theta}) > 1$

For two unbiased estimators $\hat{\theta}$ is more **efficient** than $\tilde{\theta}$ if $\text{Var}(\tilde{\theta}) \geq \text{Var}(\hat{\theta})$.

Cramér-Rao lower bound: Let $X_1, \dots, X_n$ be i.i.d. sample with pdf $f(x|\theta)$ and $T = T(X_1, \dots, X_n)$ be an unbiased estimator of θ . Then under appropriate smoothness conditions

$$\text{Var}(T) \geq \frac{1}{nI(\theta)}$$

MLE $\hat{\theta}$, $\hat{\theta}$ is unbiased.
Avar($\hat{\theta}$)

$$\text{Var}(\hat{\theta}) = \frac{1}{nI(\theta)} = \text{Avar}(\hat{\theta})$$

- If the MLE $\hat{\theta}$ is unbiased, it would be the **best (minimum variance) unbiased estimator** when n is large.

MVUE.

$$X_1, \dots, X_n \sim \text{Pois}(\lambda) \text{ i.i.d.} \quad \text{pmf. } p(x|\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Example 3.10. Let $\hat{\lambda} = \bar{X}$. $\hat{\lambda}$ is the best unbiased estimator for λ in $\text{Pois}(\lambda)$ distribution.

$$L(\lambda | \vec{x}) = \prod_{i=1}^n p(x_i | \lambda) = \lambda^{\sum x_i} \cdot \left(\prod_{i=1}^n x_i! \right)^{-1} \cdot e^{-n\lambda}$$

$$\ell(\lambda | \vec{x}) = \sum x_i \log \lambda - \log(\prod x_i!) - n\lambda$$

$$\frac{d\ell}{d\lambda} = 0 \Rightarrow \sum x_i \cdot \left(\frac{1}{\lambda} \right) - n = 0 \Rightarrow \hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

MLE estimator: $\hat{\lambda} = \bar{X}$.

$$E(\hat{\lambda}) = E(\bar{X}) = \lambda \quad \text{unbiased.}$$

$$\text{Var}(\hat{\lambda}) = \text{Var} \bar{X} = \frac{\text{Var} X_i}{n} = \frac{\lambda}{n}$$

$$\text{C.R.L.B. } \frac{1}{n I(\lambda)} = \frac{1}{I_{X_1, \dots, X_n}(\lambda)} = \frac{\lambda}{n} \quad \text{So } \hat{\lambda} \text{ is best unbiased estimator.}$$

$$I_{X_1, \dots, X_n}(\lambda) = -E\left(\frac{d^2}{d\lambda^2} \ell(\lambda | \vec{x})\right) = \frac{1}{\lambda^2} E\left(\sum x_i\right) = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$$

Exercise 3.4. Suppose $X_1, \dots, X_n$ is a random sample from $\text{EXP}(\lambda)$. We know that $E(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$. Find the MLE of the population mean $\mu = 1/\lambda$ and justify if it is the best unbiased estimator.

$$\text{MLE } \lambda, \quad \hat{\lambda} = 1/\bar{X} \quad \hat{\mu} = 1/\hat{\lambda} = \bar{X} \quad \text{by invariance}$$

$$\text{Var}(\hat{\mu}) = \frac{\text{Var}(X_i)}{n} = \frac{1/\lambda^2}{n} = \frac{1}{n\lambda^2} \quad \text{C.R.L.B.}$$

$$X_1, \dots, X_n \sim F(x|\theta), \quad \text{MMSE} \quad \hat{\theta} \quad \frac{g_1(X_1, \dots, X_n)}{g_2(X_1, \dots, X_n)} \cdot P(\theta \in (g_1, g_2))$$

$$g(X_1, \dots, X_n) \Rightarrow \theta, \quad \text{MLE} \quad \hat{\theta} \quad \frac{g_1(X_1, \dots, X_n)}{g_2(X_1, \dots, X_n)} \cdot P(\theta \in (g_1, g_2)) = 1 - \alpha$$

Confidence Interval (Interval Estimator)

Question: Is it satisfactory to report a single estimated value of θ ?
Is there an estimation method that combines both estimated value and standard error?

- Let $\mathbf{X} = (X_1, \dots, X_n)$ be a random sample with pdf $f(x|\theta)$. An interval estimator of θ with coverage probability $1 - \alpha$ is a random interval $(\hat{\theta}_L(\mathbf{X}), \hat{\theta}_U(\mathbf{X}))$.
 - $\hat{\theta}_L(\mathbf{X})$ and $\hat{\theta}_U(\mathbf{X})$ are functions of random sample $\mathbf{X}$
 - $\hat{\theta}_L(\mathbf{X}) < \hat{\theta}_U(\mathbf{X})$
 - $P\{\theta \in (\hat{\theta}_L(\mathbf{X}), \hat{\theta}_U(\mathbf{X}))\} = 1 - \alpha$. *99%, 95%, 90%*
- Called an $100(1 - \alpha)\%$ confidence interval
- $100(1 - \alpha)\%$ is referred as confidence level.
- α is a pre-specified small value, such as 0.01, 0.05 or 0.1.

$$X_1, \dots, X_n \sim N(\mu, \sigma_0^2) \quad \bar{X} \sim N(\mu, \frac{\sigma_0^2}{n}) \quad \bar{X} - \mu \sim N(0, \frac{\sigma_0^2}{n})$$

$$Q(\bar{X}, \mu) = \frac{\bar{X} - \mu}{\sigma_0/\sqrt{n}} \sim N(0, 1) \quad \text{free of } \mu.$$

Methods to construct Confidence Interval

Z-interval



$$1 - \alpha = P(a < Q(\bar{X}, \mu) < b \mid Q \sim N(0, 1))$$

$$= P(a < \frac{\bar{X} - \mu}{\sigma_0/\sqrt{n}} < b) = P(\underbrace{\bar{X} - a \cdot \sigma_0/\sqrt{n}}_{\hat{\mu}_L} < \mu < \underbrace{\bar{X} + b \cdot \sigma_0/\sqrt{n}}_{\hat{\mu}_U})$$

Exact confidence intervals: Sampling distribution of point estimator $\Rightarrow$ get a confidence interval for the parameter.

Also called pivotal method.

- exact ● Pivot. Let $Q(\mathbf{X}, \theta)$ be a function of random sample $\mathbf{X} = (X_1, \dots, X_n)$ and parameter θ . $Q(\mathbf{X}, \theta)$ is a pivot if the distribution of $Q(\mathbf{X}, \theta)$ is free of θ .
- $$P(a < Q < b) = 1 - \alpha$$
- $$= P(a^* < \theta < b^*)$$

- Example 3.11. Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ i.i.d.

- If σ^2 is known. Then $Q(\mathbf{X}, \mu) = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$ is a pivot.

- If both μ and σ^2 are unknown. Then $Q(\mathbf{X}, \mu, \sigma) = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$ is a pivot. where $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$.

- Is $(n-1)S^2/\sigma^2$ also a pivot? for σ^2 .

$a = t_{n-1}(\alpha/2), b = t_{n-1}(1-\alpha/2)$

Yes. $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. free of σ^2 .

$$1 - \alpha = P(a < Q < b)$$

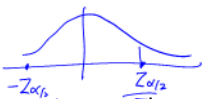
$$= P\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)S^2}{a}\right)$$

Steps for getting a confidence interval:

- 1 Find a pivot $Q(\mathbf{X}, \theta)$, whose distribution should not depend on θ .
- 2 Use the distribution of Q to find two endpoints a and b such that the probability that Q is between both points is $1 - \alpha$.
- 3 Manipulate the relation to find a lower and upper bound within which the parameter could be contained.

Example 3.11 (cont).

- When σ is known, confidence interval (Z-interval) for μ is determined by



$$1 - \alpha = P\left(-z(\alpha/2) < \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} < z(\alpha/2)\right)$$

where $z(\alpha/2)$ is upper $\alpha/2$ quantile in $N(0, 1)$. So $100(1 - \alpha)\%$ confidence interval of μ is $\bar{X} \pm z(\alpha/2)\sigma/\sqrt{n}$.

- When σ is unknown, confidence interval (t-interval) for μ is determined by

$$1 - \alpha = P\left(-t_{n-1}(\alpha/2) < \frac{\sqrt{n}(\bar{X} - \mu)}{S} < t_{n-1}(\alpha/2)\right)$$

So $100(1 - \alpha)\%$ confidence interval of μ is $\bar{X} \pm t_{n-1}(\alpha/2)S/\sqrt{n}$, where $t_{n-1}(\alpha/2)$ is the upper $\alpha/2$ quantile in t_{n-1} distribution.

$$\text{Gamma}(\alpha, \lambda), M(t) = (1 - t/\lambda)^{-\alpha}$$

$$\text{MLE of } \lambda \text{ is } \hat{\lambda} = \frac{1}{\bar{X}} = \frac{n}{\sum_{i=1}^n X_i}, \quad M_{\sum X_i}(t) = \prod_{i=1}^n M_{X_i}(t) = (1 - t/\hat{\lambda})^{-n}$$

Example 3.12. $X_1, \dots, X_n \sim \text{EXP}(\lambda)$ i.i.d. Find a $100(1 - \alpha)\%$ confidence interval for λ . $M_X(t) = (1 - t/\lambda)^{-1}$

Soln. Note that $\lambda \sum_{i=1}^n X_i \sim \text{Gamma}(n, 1)$ is a pivot.

$$\frac{M_{\lambda \sum X_i}(t)}{M_{\sum X_i}(\lambda t)} = (1 - \frac{\lambda t}{\lambda})^{-n} = (1 - t)^{-n} \propto \text{Gamma}(\frac{n}{2}, \frac{1}{2})$$

$$\text{Let } Y = \lambda \sum_{i=1}^n X_i \quad Q(\bar{X}, \lambda) \quad \chi_n^2 = \text{Gamma}(\frac{n}{2}, \frac{1}{2})$$

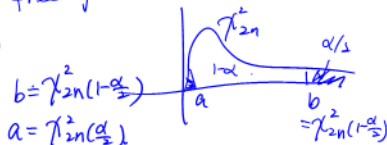
$$M_Y(t) = M_{\lambda \sum X_i}(t) = (1 - 2t)^{-n/2} = (1 - \frac{t}{1/2})^{-n/2}$$

$$\Rightarrow Y \sim \text{Gamma}(\frac{n}{2}, \frac{1}{2}) = \chi_{2n}^2 \text{ free of } \lambda$$

$$1 - \alpha = P(a < Y < b \mid Y \sim \chi_{2n}^2)$$

$$= P(a < \lambda \sum X_i < b)$$

$$= P(\frac{a}{\sum X_i} < \lambda < \frac{b}{\sum X_i})$$



Exercise 3.5. $X_1, \dots, X_n \sim N(0, \sigma^2)$ i.i.d. Find a $100(1 - \alpha)\%$ confidence interval for σ^2 .

$$X_i \sim F(x|\theta). \text{ MLE of } \theta, \hat{\theta}.$$

Asymptotic confidence intervals: when exact distributed pivot is hard to find and sample size n is big, we can construct approximate C.I. based on the asymptotic normality of MLE.

● Recall.

$$\sqrt{nI(\theta_0)}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, 1), n \rightarrow \infty.$$

Then

$$\hat{\theta} \overset{\alpha}{\sim} N(\theta, \frac{1}{nI(\theta)}).$$

$$\bar{X} \sim N(\mu, \frac{\sigma_0^2}{n})$$

Approx $1-\alpha$ C.I. of θ : $\hat{\theta} \pm z(\alpha/2) / \sqrt{nI(\hat{\theta})}$

$1-\alpha$ C.I. of μ :

$$1-\alpha \approx P\left(-z(\alpha/2) < \sqrt{nI(\hat{\theta})}(\hat{\theta} - \theta_0) < z(\alpha/2)\right).$$

$$\bar{X} \pm z(\alpha/2) \cdot \frac{\sigma_0}{\sqrt{n}}$$

- Therefore, an approximate $100(1-\alpha)\%$ confidence interval for μ is

$$\hat{\theta} \pm z(\alpha/2) \frac{1}{\sqrt{nI(\hat{\theta})}},$$

where $z(\alpha/2)$ is upper $\alpha/2$ quantile in $N(0, 1)$.

